ACADEMIC PROFILE

Curriculum vitae

Iulian-Teodor Voicilă, PhD · iulian.voicila@upb.ro · ORCID 0000-0002-6033-2934

Department of Measurements, Electrical Apparatus and Static Converters, Faculty of Electrical Engineering, National University of Science and Technology POLITEHNICA Bucharest.

Experience

September 2026 - present

Teaching Assistant

National University of Science and Technology POLITEHNICA Bucharest. Department of Measurements, Electrical Apparatus and Static Converters. Lectures and laboratories in Data Transmissions and Protocols and in Measurements and Instrumentation.

May - September 2026

Specialist Engineer

National University of Science and Technology POLITEHNICA Bucharest. Department of Measurements, Electrical Apparatus and Static Converters. Operation of high-precision laboratory equipment, calibration and metrological verification of power-quality measurement instruments, and traceable testing to current standards in electrical metrology and power quality.

October 2025 - May 2026

Associate Teaching Staff

National University of Science and Technology POLITEHNICA Bucharest. Department of Measurements, Electrical Apparatus and Static Converters.

November 2021 - September 2025

Teaching Assistant

National University of Science and Technology POLITEHNICA Bucharest. Department of Measurements, Electrical Apparatus and Static Converters. Laboratories in electrical measurements, measurement systems and biomedical sensors. Secretary of bachelor and master examination committees, supervision of bachelor student groups and of student scientific session projects, technical support for the ATEE conference.

May 2019 - August 2022

Functional Safety Engineer, automotive

Expleo Romania, Bucharest. Embedded E/E, System Functional Safety. Safety analyses for vehicle systems and inter-system studies under ISO 26262 (HARA, FMEA/DFMEA, FTA, DFA, safety concepts, validation plans), CAN network and architecture analysis, ECU testing. Internal trainer for IBM DOORS.

June - September 2018

Engineering Intern

ICPE SAERP SA, Bucharest. Testing and assembly of electrical equipment in electric drives and static converters.

Education

2021 - 2025

PhD in Electrical Engineering

National University of Science and Technology POLITEHNICA Bucharest · Doctoral School of Electrical Engineering. Thesis: Intelligent sorting system for end-of-life Li-ion batteries from the automotive industry. Supervisor: Prof. George-Călin Seriță.

2019 - 2021

MSc · Power Electronics and Intelligent Electric Drives

Politehnica University of Bucharest · Faculty of Electrical Engineering. Dissertation: Design and implementation of a selection system for automotive batteries.

2015 - 2019

BSc (Eng.) · Power Electronics and Electric Drives

Politehnica University of Bucharest · Faculty of Electrical Engineering. Diploma project: Virtual instrument for generating and measuring periodic signals.

2020 - 2022

Teacher training programme, levels I and II

Valahia University of Târgoviște · Teacher Training Department

Research projects

From February 2025

EVOSST, Evaluation and Valorization of Social Services for a Sustainable Transition

Research engineer (team member). Horizon Europe · Research and Innovation Action, grant 101178789.

January 2024 - January 2026

TwinEU, Digital Twin for Europe

Research engineer (team member). Horizon Europe · Innovation Action, grant 101136119.

October 2021 - September 2024

ELECTRON, rEsilient and seLf-healed EleCTRical pOwer Nanogrid

Research engineer (team member). EU Horizon 2020 · Innovation Action, grant 101021936.

July 2020 - September 2021

STOCMAN, Intelligent management system for improving the efficiency of energy storage systems based on Pb-acid and Li-ion batteries

Research engineer (team member). POC 2014-2020 · Action 1.2.1.

Research output

9 journal articles and 10 conference papers, 3 laboratory books, 2 Romanian patent applications. Research topics: second-life battery screening, state-of-health and OCV estimation, power converters for battery testing, measurement systems.

Browse the publication catalogue →

Software and open data

2026

MEE learning application for Electrical and Electronic Measurements

Author. Bilingual (Romanian/English) app for Android and the web, open source under Apache-2.0. Available on Google Play; source code at github.com/BulyK47/mee-app; archived on Zenodo, DOI 10.5281/zenodo.23105028.

2026

24-Hour Raw OCV Relaxation Dataset for Rechargeable Batteries with Different Chemistries

Co-author. IEEE DataPort, CC BY 4.0, DOI 10.21227/651q-8v82.

2026

Battery Charge-Discharge Voltage Curves

Co-author. IEEE DataPort, CC BY 4.0, DOI 10.21227/cm0f-jg66. Both datasets are mirrored at github.com/BulyK47/battery-datasets.

Teaching

Lectures and laboratories in Data Transmissions and Protocols and in Measurements and Instrumentation. Laboratories in Electrical and Electronic Measurements (Romanian and English programmes), Measurement of Electrical and Non-Electrical Quantities, Measurement Systems and Biomedical Sensors. Erasmus+ teaching mobility, University of Western Macedonia, February 2025.

Languages

Romanian (native), English (C1), German (A1), French (A1).